
Short-Term Weather Forecasting for Precision Agriculture in Jammu and Kashmir: A Deep-Learning Approach

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14.1 INTRODUCTION

Agricultural production serves as a primary source of economic development in any country, and weather forecasting (WF) directly or indirectly influences the yearly profitability or loss of agricultural production [1]. WF enables farmers to anticipate and prevent potential damages caused by intense weather phenomena, including reduced yields, misirrigation, erroneous crop selection, and inaccurate harvesting times. Unpredictable weather changes can significantly hinder agricultural progress for an extended period of time. The capacity to anticipate approaching snowfall, rainfall, heatwaves, and floods aids farmers in strategizing and subduing the devastating impacts of such events.

With the advancement of artificial intelligence (AI) in WF, the reliability and accuracy of predictions has significantly improved. Data-driven AI algorithms like neural networks (NN) can be employed to analyse historical weather data and forecast future trends and patterns in weather. These models also improve the accuracy of short-term weather predictions, aiding agriculturists to take informed decisions. Conversely, the performance of statistical techniques such as Autoregressive Moving Average (ARMA) and its variations relies on both the quality of the data and their capability to accurately represent time-series (TS) data exhibiting non-stationary patterns across different seasons [2]. The data-driven modeling systems can be

employed to decrease the computational requirements of Numerical Weather Prediction (NWP) while also accounting for the uncertainty present in observations and system noise [3]. This can be ascribed to the capacity of data-driven models to estimate the intrinsic patterns and dynamics of TS data, even without prior knowledge of the parameters. Specifically, Artificial Neural Networks (ANNs) can be employed for this objective owing to their adaptive characteristics and capacity to learn from previous knowledge [4]. These characteristics make ANNs appealing in diverse fields, particularly when dealing with intricate nonlinear phenomena present in datasets such as weather datasets. Different categories of NN, such as feedforward NN, NN employing back propagation, Recurrent Neural Networks (RNNs), and Long Short-Term Memory (LSTM), adopt a distinct approach to prediction, each offering a unique method for achieving accurate results.

To effectively handle weather data, RNN and LSTM models are considered superior choices due to their ability to efficiently process TS data [5]. Conventional NNs lack the necessary power to capture long-term dependencies. As the number of layers increases, they are prone to encountering the vanishing gradient problem. On the other hand, Recurrent Neural Networks (RNNs) are well-suited for handling TS data with diverse temporal scales, allowing them to store long-term information. For the training of RNN, a TS dataset consisting of daily records of minimum temperature (min temp), maximum temperature (max temp), and rainfall values has been acquired from the Indian Meteorological Department (IMD). This study aims to establish a predictive modeling framework for short-term WF in order to assist farmers in efficient agricultural resource management and plan their farming activities ahead of time. The model will utilize past values within a defined time step as input to forecast min temp, max temp, and rainfall values for the shorter number of days.

14.1.1 Contributions

The primary contributions of this study are summarized as follows:

- To reduce the direct and indirect economic load caused by to reliance of farmers on erroneous predictions of weather
- To proactively reduce potential damages in the agricultural sector resulting from severe weather situations
- To construct a deep-learning (DL) model specifically designed for short-term WF using TS data

- To generate precise weather predictions within a limited timeframe by utilizing localized weather data
- To construct a WF model that demonstrates strong adaptability to evolving trends in weather data and effectively handles extreme weather events while maintaining consistent performance

14.2 RELATED WORK

Various techniques based on both statistical methods and data-driven models have been devised over time to forecast weather conditions by predicting one or more weather-related parameters. Researchers [6] conducted a study to predict rainfall intensity using daily temperature, humidity, wind direction, wind speed, and cloud speed. The prediction models were built using Decision Tree (DT), Support Vector Regression (SVR), and Random Forest (RF), achieving an R^2 score of 0.806, 0.900, and 0.980, respectively. In another study, researchers [7] trained different machine-learning models for rainfall prediction in order to support precision agriculture in Sri Lanka. The models were trained using attributes like rain gauge, relative humidity, average temperature, wind speed, wind direction, solar radiation, and ozone concentration specific to the region of Sri Lanka. It was observed that RF performed best with an accuracy of 89.16%. In their work, researchers [8] combined machine-learning techniques with data mining techniques to develop a rainfall prediction model, and it was observed that Autoregressive Integrated Moving Average (ARIMA) models and NNs provide the best accuracy of 70.5%. In the last decade, RNNs have attracted substantial attention and seen significant progress because of their strong and influential modeling capabilities [9]. In a study, researchers [10] employed daily discharge and rainfall to forecast floods. The research findings suggested the potential use of LSTM, which is a variant of RNN, for the development and administration of real-time flood warning systems. In their experimental study, researchers [1] concluded that the LSTM model outperforms the ARIMA model in predicting minimum and maximum temperature by a root mean squared error (RMSE) score of 1.04. The research carried out by Gumaste and Kadam [11] Placeholder Textemployed the Genetic Algorithm (GA) and Fast Fourier Transform (FFT) for future weather prediction using present data [11]. The models were subsequently deployed through mobile

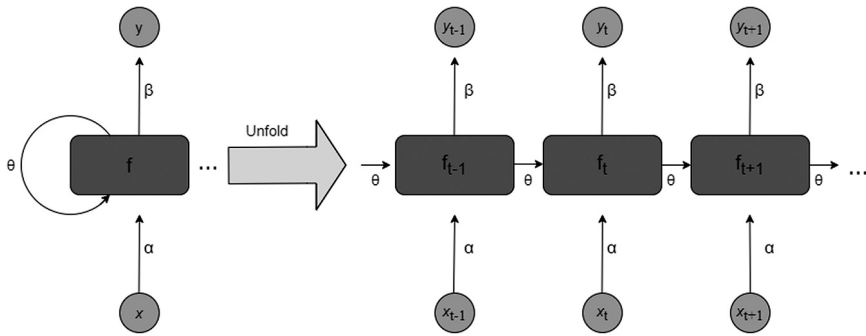
apps to assist farmers in planning their pre- and post-agricultural tasks. To achieve precise weather forecasts for the Dellangu village of Indonesia, researchers developed a short-term weather prediction model utilizing ANNs to help farmers deal with their rice field problems [12]. Another research conducted a comparative analysis between ten base machine-learning regressors and an ensemble model [13]. The analysis suggested that the ensemble model performed better than base models for predicting rainfall, relative humidity, and maximum and minimum temperature. In order to reduce the cost of agricultural frost damage, researchers trained machine-learning models like the convolutional neural network, deep neural network, and RF to predict frost events [14]. The research established that soil temperature is a crucial factor when making predictions beyond 24 hours, whereas for shorter-term forecasts, most of the relevant information is derived from other temperature-related parameters.

14.3 PROPOSED WORK

This study presents an innovative and lightweight WF model specifically designed for accurate short-term predictions of temperature and rainfall in Jammu and Kashmir (J&K). The model is designed to be compatible with stand-alone PCs and can be readily implemented in the geographical region J&K. The proposed system utilizes an advanced DL technique known as RNN that is suitable for sequential data and can enhance the accuracy of the predictions. The following subsection provides an elaborate explanation of the RNN model.

14.3.1 Recurrent Neural Network

A traditional RNN [15] can be thought of as a variation of a feedforward NN that possesses an internal memory component to process the input sequences. RNNs have the ability to learn dependencies from previous terms and predict cumulative measures from variable-length sequences, which include a time component. RNNs are particularly useful for developing prediction models using long-term TS datasets, which can be useful in WF. Unlike traditional deep NN that assume inputs and outputs are unrelated to one another, RNNs generate outputs that are influenced by preceding elements within the sequence. For that, RNNs utilize feedback

**FIGURE 14.1**

Working of RNN.

loops to leverage the output from one time step as input for the subsequent step. By maintaining information in a sequential manner, the input of the RNN at each time step incorporates both the data from the current time step in the sequence and the preceding information passed through the feedback loop. The recurrent nature of an RNN stems from its consistent application of the same function to compute the output for each input, with the output being influenced by previous calculations.

Figure 14.1 illustrates a basic RNN architecture in which x_t to x_{t+1} represent the inputs for each sequence, and y_t to y_{t+1} represent the corresponding outputs produced for each sequence. It is evident that all the inputs are interconnected, with each connection denoted by a single RNN cell. The terms f_{t-1} to f_{t+1} refer to hidden states, and θ represents the weight of hidden states; α signifies the weight of the current input state; β represents the weight at the output state. In contrast to traditional feedforward NNs where weights are distinct for each node, RNNs utilize shared parameters across the layers, implying that within each layer of the RNN, the same weight parameter is applied. However, these shared weights are still updated during the backpropagation and gradient descent processes to enable effective reinforcement learning. The functioning of RNN can be explained in the following steps:

For a sequence of data points x , denoted as $x = [x_{t-1}, x_t, x_{t+1} \dots]$, where x_t represents the input at a specific time step “ t ” within the sequence:

- The hidden state at time step “ t ”, denoted as f_t , is calculated by combining the current input x_t with the previous hidden state f_{t-1} . This computation is expressed as

$$f_t = \sigma(\alpha x_t + \theta f_{t-1}),$$

where σ represents the activation function applied to the sum of weighted inputs; α represents the weight matrix for the connections from the input to the hidden state; θ represents the weight matrix for the connections within the hidden state.

- The output y_t at time step “ t ” can be computed using the hidden state:

$$y_t = \sigma(\beta f_t)$$

where β is the weight matrix for the hidden-to-output connections.

The activation function is a mathematical operation used to incorporate non-linearity into NN, allowing them to effectively learn intricate data patterns and make nonlinear changes to the input data. In RNNs, the activation function is exclusively applied to the hidden layers since the input layer solely holds the input data and does not undergo any calculations. Additionally, the output of the RNN, which represents the network’s prediction or generated output at each time step, can also undergo an activation function to convert the raw output into a preferred range or format based on the specific task or problem being tackled. Without the presence of nonlinear activation functions, a neural network with multiple hidden layers would essentially resemble a large-scale linear regression model and would be unable to effectively learn intricate and complex patterns from real-world data. Commonly used activation functions are Rectified Linear Unit (ReLU), tanh, Sigmoid, and Softmax.

14.3.1.1 Backpropagation through Time (BPTT)

BPTT [16] is an iterative training method utilized specifically for RNNs. It extends the standard backpropagation algorithm, commonly employed for training feedforward NN, and incorporates the temporal aspect of RNNs by allowing gradients to propagate backward across time, considering the sequential nature of the TS data. In RNN, the hidden state of the network at each time step is influenced by both the current input and the output from the previous hidden state. This gives rise to a chain of dependencies that stretches back in time, allowing BPTT to unfold the RNN over time. A simplified breakdown of the steps involved in BPTT is given next:

For a sequence of data points x , denoted as $x = [x_{t-1}, x_t, x_{t+1} \dots]$, where x_t represents the input at a specific time step “ t ” within the sequence.

- The hidden state at time step “ t ”, denoted as f_t , is calculated by combining the current input x_t with the previous hidden state f_{t-1} . This computation is expressed as

$$f_t = \sigma(\alpha x_t + \theta f_{t-1}),$$

Where σ , α , and θ have the usual meanings.

- The loss function, such as mean squared error (MSE) or cross-entropy loss, is applied to calculate the loss at each time step. We can represent the loss at a specific time step “ t ” as l_t .

The gradients of the loss with respect to the parameters of the network are calculated from the final time step. Subsequently, these gradients are propagated in a backward direction across previous time steps. The gradient at time step “ t ,” which signifies the impact of the loss on the hidden state at “ t ” can be represented as:

$$\frac{\partial l_t}{\partial f_t}$$

- The gradients are accumulated at each step as they propagate through time, ensuring that dependencies between different time steps are captured effectively. Accumulated gradients are represented as follows.

$$\frac{\partial l}{\partial f_t}$$

- The parameters of the network are updated by utilizing optimization algorithms such as Adam, SGD and RMSprop. The parameter update for each weight can be computed as

$$\Delta w = -\eta * \sum \frac{\partial l}{\partial w},$$

where Δw is the update in weight, and η is the learning rate.

14.4 EXPERIMENTS

Figure 14.2 illustrates the comprehensive process of modeling in detail. The experimentation initiates with the data collection and data preprocessing phase, which is elaborated in the subsequent section. Following that, the process encompasses model configuration, training, comparison, and evaluation. The final step of the experimentation involves assessing and contrasting the performance of the trained models.

14.4.1 Data Collection

This study utilizes real meteorological data obtained from the IMD website [17], specifically from weather stations in Jammu, Srinagar, and Leh. The dataset for each weather station consists of a daily TS comprising 8,401 data points, spanning from January 1, 2010, to December 31, 2022. The dataset includes various weather variables, including rainfall, minimum temperature, and maximum temperature. Rainfall values are measured

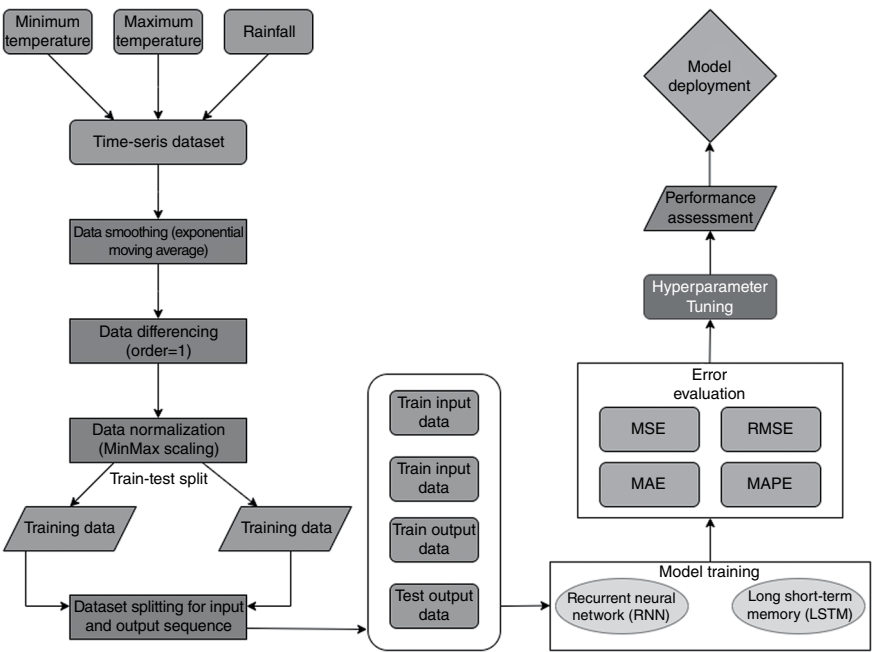


FIGURE 14.2
Illustration of modeling.

in millimeters (mm), while temperature variables are recorded in Celsius. These specific meteorological features are chosen because they provide valuable insights into the current weather conditions at a particular location and time. The selection of these features aims to capture the state of the weather accurately. In this study, these three meteorological features are utilized as input variables for predicting temperature.

14.4.2 Data Preprocessing

The experiment conducted on real data necessitated several preprocessing steps to accurately reflect the performance of the model. In the beginning, smoothing of data has been performed to reduce noise and fluctuations and make underlying patterns and trends more evident. Data has been smoothened using the Exponential Moving Average (EMA) technique. EMA [18] is the weighted average value where weights are decreased gradually such that more importance is given to recent data points as compared to the historical ones, or vice versa. EMA changes at a faster rate and is more sensitive to the data points. Mathematically, the EMA of a data point can be calculated using Equation 14.1.

$$\text{EMA}_t = \begin{cases} x_0 & t = 0 \\ \alpha x_t + (1 - \alpha) \text{EMA}_{t-1} & t > 0 \end{cases} \quad (14.1)$$

where α is a smoothing factor, and its value lies between 0 and 1. It represents the weight applied to the very recent period. For this study, the value of α has been kept at 0.5. To make the TS stationary, differencing of order one has been performed on the weather dataset. Differencing entails computing the differences between consecutive observations by subtracting the previous value from the current value. The purpose of differencing is to eliminate the underlying trend and isolate the specific changes or fluctuations present in the data. For the final stage of preprocessing, data has been normalized using the MinMax scaling technique because the values of different variables in the dataset have different units and lie in multiple ranges. Mathematically, MinMax scaling is represented by Equation 14.2.

$$X_t = \frac{X_t - X_{\min}}{X_{\max} - X_{\min}}, \quad (14.2)$$

where X_t represents the data point at time “ t ” in TS and $X_{\max}$ and $X_{\min}$ represent maximum and minimum data points in the sequence, respectively.

Normalized data allows models to generalize well on new and unseen data. Additionally, activation functions in RNN and LSTM neurons work more effectively on normalized data. The dataset has been divided into a training set and a testing set, with proportions of 0.8 and 0.2, respectively. Each weather station’s training dataset consists of 6,720 rows containing weather data from January 1, 2000, to May 25, 2018, while the testing dataset contains 1,680 rows spanning from May 26, 2018, to December 31, 2022. Input and output sequences were obtained using the moving window algorithm and a window size of seven. Both input and output sequences contain three features.

14.4.3 Model Configuration and Training

The proposed models have been trained using two distinct configurations: Single Input Single Output (SISO) and Multiple Input Multiple Output (MIMO) [19]. SISO is a forecasting arrangement wherein a single input variable is fed into the network to forecast the future values of the same variable as output. On the other hand, in MIMO, multiple variables are fed into the network in order to predict the same variables as output. Using this configuration, only one model is required per weather station to predict all the future values, while in SISO, three different models are needed for each weather station to forecast three weather parameters. Therefore, three RNN models were trained and evaluated using the MIMO configuration for three distinct weather stations. In addition, a total of nine RNN models were both trained and tested in SISO configuration to forecast three temperature variables across three weather stations. Both configurations have been trained on different sets of hyperparameters of RNN. Hyperparameters are specific values that must be defined prior to model training and are not subject to learning or adjustment during the model training process. The hyperparameters tuned in this study include the number of epochs, batch size, optimization algorithms, and activation functions. The technique chosen for tuning hyperparameters is GridSearchCV, which automates the process of trying various hyperparameter configurations. It thoroughly explores a pre-defined grid of hyperparameter values to identify the combination that produces the highest model performance. For the RNN model, the number of epochs defines how many times the model will be trained

on an entire dataset [20–23]. During each epoch, the model goes through one forward pass (input to output) and one backward pass (adjusting weights based on gradients) for all training samples. Using a smaller number of epochs leads to increased error loss, while a larger number of epochs may cause overfitting. The models have been trained for a range of 50–300 epochs. The batch size refers to the number of training samples processed together by the model in one forward and backward pass, allowing the model to update the weights after processing a group of samples. Using a larger batch size during model training comes with higher computational costs, while a smaller batch size introduces noise in the model. Hence, training experiments have been conducted with batch sizes of 16, 32, 64, and 128 to find the optimal configuration. The choice of the optimization algorithm impacts both the speed at which the model reaches its optimal solution and its ability to generalize effectively to unseen data. In order to identify the most suitable optimization algorithm, the models underwent training using Adam, SGD, and RMSprop optimizers. For selecting the optimal activation function, models have been trained on tanh, ReLU, and sigmoid activation functions.

14.4.4 Model Evaluation

The models have been repeatedly trained on training data and evaluated on testing data. For evaluating TS regression models, the commonly used metrics are mean absolute error (MAE), MSE, RMSE, and mean absolute percentage error (MAPE), which are calculated by using Equations 14.3–14.6 [24–26].

$$\text{MAE} = \sum_{i=1}^N \left| y_t - \hat{y}_t \right| \quad (14.3)$$

$$\text{MSE} = \sum_{i=1}^N \left(y_t - \hat{y}_t \right)^2 \quad (14.4)$$

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^N \left(y_t - \hat{y}_t \right)^2}{n}} \quad (14.5)$$

$$\text{MAPE} = \sum_{i=1}^N \frac{|y_t - \hat{y}_i|}{y_i} \tag{14.6}$$

where y_i and $\hat{y}_t$ are the actual and predicted values of temperature variables at time step “ t .”

14.5 RESULTS AND DISCUSSION

As mentioned in Section 14.3, the RNN model has been trained using different sets of hyperparameters and tested using various evaluation metrics. For each station, it has been observed that the RNN model performs noticeably better in the SISO configuration than the MIMO configuration. The values of different accuracy metrics for RNN models are listed in Table 14.1 for all weather stations. Tables 14.2–14.4 list the accuracy metrics of the RNN model in SISO configuration for Jammu, Srinagar, and Leh, respectively. These findings indicate that the variables present in weather data are relatively univariate and do not possess strong interdependencies between them. Such relationships can be adequately represented using a single input variable to forecast a single output variable [27–29].

14.6 CONCLUSION AND FUTURE SCOPE

This study highlights the importance of choosing regional data in building a reliable WF model and tackling the WF problem by focusing on

TABLE 14.1
Model Performance in MIMO Configuration for all the Stations

Metrics	Corresponding Value for Jammu Station	Corresponding Value for Srinagar Station	Corresponding Value for Leh Station
MAE	0.0636	0.1048	0.8543
MSE	0.0401	0.0455	0.0411
RMSE	0.1011	0.1555	0.1344
MAPE	0.3990	0.4398	0.3229

TABLE 14.2

Model Performance in SISO Configuration for Jammu Station

Temperature Variables	$T_{\min}$	$T_{\max}$	Rainfall
MAE	0.0751	0.0690	0.0124
MSE	0.0095	0.0095	0.0257
RMSE	0.0977	0.0091	0.0006
MAPE	0.1863	0.1683	0.1229

TABLE 14.3

Model Performance in SISO Configuration for Srinagar Station

Temperature Variables	$T_{\min}$	$T_{\max}$	Rainfall
MAE	0.0658	0.0696	0.0210
MSE	0.0073	0.0926	0.0018
RMSE	0.0854	0.0085	0.0429
MAPE	0.1436	0.1696	0.1368

TABLE 14.4

Model Performance in SISO Configuration for Leh Station

Temperature Variables	$T_{\min}$	$T_{\max}$	Rainfall
MAE	0.0666	0.0085	0.0640
MSE	0.0075	0.0015	0.0078
RMSE	0.0866	0.0393	0.0887
MAPE	0.1547	0.0588	0.1623

achieving precise forecasts without requiring substantial computational resources, in contrast to NWP methods. To resolve this, a lightweight DL-based prediction model has been proposed to ensure accurate short-term WF. The suggested model is capable of operating independently on a single computer unit and can be readily implemented in the geographical area of J&K. In future, more variables like wind speed, humidity, and pressure could be incorporated into the process of model training to provide a more diverse weather forecast. The results also showed that SISO performed better than MIMO configurations; however, that would require three different models to predict three weather parameters as output. In future experiments, MIMO configurations could be further optimized to enhance both performance and efficiency simultaneously.

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